Assignment 2

Group 85

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Part 1: Analyzing the Mincerian wage model

Exercise 1.1

We start by estimating the model.

```
 Intercept  Education Experience
     1.037      0.118      0.022
```

As the estimated OLS coefficients are small, we can conclude that one unit change in a regressor $x_{k,i}$ translates to $\beta_k \times 100\%$ change in y_i. In other words, each additional unit of education increases wages by 11.8%, whereas each additional unit of experience increases wages only by 2.2%.

Exercise 1.2

The correlation coefficient can be measured as

$$r_{\hat{\beta}_2,\hat{\beta}_3} = \frac{Cov(\hat{\beta}_2, \hat{\beta}_3)}{\sqrt{Var(\hat{\beta}_2)Var(\hat{\beta}_3)}}.$$

Computing the expression in R yields the following result.

```
[1] -0.056
```

Exercise 1.3

Part a)

We test non-linearity using the regression specification error test "RESET" of Ramsey. The hypotheses are the following: $H_0 : \gamma = 0$ vs $H_a : \gamma \neq 0$ for test equation which is $y_i = x_i'\beta + \gamma\hat{y}_i^2 + \epsilon_i$. The regular estimates rely on the classical OLS variance formula, $\sigma^2(X'X)^{-1}$. The robust estimates, however, use the Eicker-Huber-White 'sandwich' form, $(X'X)^{-1}(X'\Omega X)(X'X)^{-1}$, providing a heteroskedasticity-consistent covariance matrix. After computing the t-test, under H_0, for both standard error variations, we get the following results:

```
t-test critical values at different significance levels:

at 1%  | critical value: 2.5858

at 5%  | critical value: 1.9648

at 10% | critical value: 1.6479

Regular test: t-test value: 1.2425  | p-value: 0.21463

Robust test: t-test value: 1.1707  | p-value: 0.2423
```

These results conclude that there is no significant evidence to reject the null hypothesis for both calculations - with regular standard errors and with robust standard errors - as t-test values are not in the rejection region and p-values are greater than 1%, 5% or 10% significances.

Part b)

We want to test for the hypothesis $H_0 : \gamma_1 = \gamma_2 = \gamma_3 = 0$ vs H_a : at least one $\gamma_j \neq 0, j = 1, 2, 3$ for the test equation $y_i = x_i'\beta + \gamma_j \Sigma_{j=1}^{3} \hat{y}_i^{1+j} + \epsilon_i$. After computing Wald test statistics (df = 3), using regular and robust to heteroscedasicity SE's, we get the following results for 1%, 5% and 10% significance levels:

```
Wald-test critical values at different significance levels (df=3):

at 1%  | critical value: 11.3449

at 5%  | critical value: 7.8147

at 10% | critical value: 6.2514

Regular test: Wald value: 3.3901  | p-value: 0.3353

Robust test:  Wald value: 5.4085  | p-value: 0.14422
```

Again, we conclude that there is not enough evidence to reject the null hypothesis as p-values are above the significance levels and Wald-statistics for both - regular and robust - SE's are not in the rejection region.

Exercise 1.4

We want to test for heteroscedasticity, where we will use Breusch-Pagan test and use the regressors of the original model as z-variables. After following the three steps, the results we get are:

```
Breusch-Pagan Test (LM Statistic):

at 1%  | critical value: 9.2103

at 5%  | critical value: 5.9915
```

```
at 10% | critical value: 4.6052

Breusch-Pagan Test Results:

LM statistic:  13.6373  | p-value: 0.00109
```

Since this p-value (0.00109 or 0.109%) is strictly less than the significance levels of 1%, 5%, and 10%, we reject the null hypothesis H_0 of homoskedasticity. By rejecting the null hypothesis, we can conclude that the variance of the error terms is not constant across the observations.

While our current test, using the original regressors educ and exp, confirms the presence of heteroskedasticity, the conclusion is technically sensitive to the choice of z-variables. If the true heteroskedasticity is non-linear or driven by excluded factors, a test using only linear z-variables might lack the power to detect it. However, given our very low p-value, it is highly likely that heteroskedasticity is a robust feature of this dataset regardless of the specific z-variables chosen.

Part 2: Simulating the RESET test and Monte Carlo

```
[1] "We are using Dataset bankwage_data.csv"

Part 2 setup: n = 500, beta4 = 0.0040, x4 = exp^2
```

We generate data from $y_i = 1 + 0.1\,educ_i + 0.025\,exp_i + 0.004\,exp_i^2 + \varepsilon_i$ with $\varepsilon_i \sim N(0,1)$ and $n = 500$.

Exercise 2.1: One sample

```r
set.seed(42)
eps  = rnorm(n, 0, sigm0)                        # generate disturbances N(0, sigma^2)
ysim = Xdgp %*% beta0 + eps                      # together with X, your dataset is complete

# Estimate the misspecified model: y = b1 + b2*educ + b3*exp + error, considering only linear features

X = cbind(rep(1, n), educ, exp)
k = ncol(X)

XtX       = t(X) %*% X
XXinv     = solve(XtX)
diagXXinv = diag(XXinv)
XXinvX    = XXinv %*% t(X)                        # part of estimator that remains the same

bhat = XXinvX %*% ysim
yhat = X %*% bhat
e    = ysim - X %*% bhat
s2   = sum(e * e) / (n - k)
seb  = sqrt(s2 * diagXXinv)

cat("Estimated coefficients (misspecified model):\n")
```

```
Estimated coefficients (misspecified model):
```

```r
results = cbind(bhat, seb)
colnames(results) = c("b_hat", "se(b)")
rownames(results) = c("constant  ", "educ", "exp")
print(results, digits = 5)
```

```
              b_hat     se(b)
constant   0.967097 0.2574502
educ       0.084587 0.0183054
exp        0.106678 0.0065623
```

RESET test with $p = 3$

```r
# Unrestricted model: add yhat^2, yhat^3, yhat^4
Xu     = cbind(X, yhat^2, yhat^3, yhat^4)
ku     = ncol(Xu)                                  # 6 regressors in unrestricted model
bhat_u = solve(t(Xu) %*% Xu, t(Xu) %*% ysim)
eu     = ysim - Xu %*% bhat_u
ssrR   = sum(e * e)                                # RSS restricted (original model)
ssrU   = sum(eu * eu)                              # RSS unrestricted (augmented model)
q      = 3                                         # number of restrictions

Fstat  = ((ssrR - ssrU) / q) / (ssrU / (n - ku))

# Robust Wald
XXinvU = solve(t(Xu) %*% Xu)
XXinvU = solve(t(Xu) %*% Xu)
ku = ncol(Xu)
Sigma_hat = matrix(0, ku, ku)
for (i in 1:nrow(Xu)) {
  xi = Xu[i, ]
  Sigma_hat = Sigma_hat +  e[i]^2* xi %*% t(xi)
}
Var_rob   = XXinvU %*% Sigma_hat %*% XXinvU
R      = cbind(matrix(0, q, 3), diag(q))    # We define a constraints such that last three regressors a
                                              # r = (0,0,0)', so ommitted
Rb_hat     = R %*% bhat_u
Wrob   = (t(Rb_hat) %*% solve(R %*% Var_rob %*% t(R)) %*% Rb_hat)

cat("RESET test (p=3) on simulated data:\n")
```

```
RESET test (p=3) on simulated data:
```

```r
cat(sprintf("  F-stat (regular) = %.4f   p-value = %.4f\n",
            Fstat, 1 - pf(Fstat, q, n - ku)))
```

```
  F-stat (regular) = 1.0582   p-value = 0.3665
```

```r
cat(sprintf("  Wald (robust)  = %.4f   p-value = %.4f\n",
            Wrob, pchisq(Wrob, df = 3, lower.tail = FALSE)))
```

```
  Wald (robust)  = 3.3781   p-value = 0.3369
```

We see that using both the simple F-test and robust wald test, the P-value> 0.05, meaning that we reject the null hypothesis: 'The model has no ommitted variables'

Exercise 2.2: 10,000 simulations

```
nsim = 10000

F_cv_1  = qf(0.99, 3, n - 6)                      # critical values F(3, n-6)
F_cv_5  = qf(0.95, 3, n - 6)
F_cv_10 = qf(0.90, 3, n - 6)
W_cv_1  = qchisq(0.99, 3)                       # critical values Chisquare(3)
W_cv_5  = qchisq(0.95, 3)
W_cv_10 = qchisq(0.90, 3)


nrej1 = 0; nrej5 = 0; nrej10 = 0    #Normal rejects
Rnrej1 = 0; Rnrej5 = 0; Rnrej10 = 0  # Robust rejects

set.seed(42)
for (i in 1:nsim) {
  eps  = rnorm(n, 0, sigm0)                     # generate disturbances N(0, sigma^2)
  ysim = Xdgp %*% beta0 + eps                   # data from the TRUE model
  bhat   = XXinvX %*% ysim
  yhat   = X %*% bhat             # OLS on misspecified model
  e      = ysim - yhat
  ssrR   = sum(e * e)


  Xu     = cbind(X, yhat^2, yhat^3, yhat^4)    # RESET augmentation
  bhat_u = solve(t(Xu) %*% Xu, t(Xu) %*% ysim)
  eu     = ysim - Xu %*% bhat_u
  ssrU   = sum(eu * eu)


  #Simple F test
  Fstat  = ((ssrR - ssrU) / 3) / (ssrU / (n - 6))

  if (Fstat > F_cv_1)  nrej1  = nrej1  + 1
  if (Fstat > F_cv_5)  nrej5  = nrej5  + 1
  if (Fstat > F_cv_10) nrej10 = nrej10 + 1

   #Robust test
  XXinvU = solve(t(Xu) %*% Xu)
  XXinvU = solve(t(Xu) %*% Xu)
  ku = ncol(Xu)
  Sigma_hat = matrix(0, ku, ku)
  for (i in 1:nrow(Xu)) {
    xi = Xu[i, ]
    Sigma_hat = Sigma_hat +  eu[i]^2* xi %*% t(xi)
  }
  Var_rob   = XXinvU %*% Sigma_hat %*% XXinvU
  R      = cbind(matrix(0, q, 3), diag(q))    # We define a constraints such that last three regressors
                                                # r = (0,0,0)', so omitted
  Rb_hat     = R %*% bhat_u
  Wrob   = (t(Rb_hat) %*% solve(R %*% Var_rob %*% t(R)) %*% Rb_hat)
```

```r
  if (Wrob > W_cv_1)  Rnrej1  = Rnrej1  + 1
  if (Wrob > W_cv_5)  Rnrej5  = Rnrej5  + 1
  if (Wrob > W_cv_10) Rnrej10 = Rnrej10 + 1

}

cat("Rejection rates over 10000 simulations:\n")
```

```
Rejection rates over 10000 simulations:
```

```r
cat(sprintf("  1%%:  %.2f%%\n", nrej1  / nsim * 100))
```

```
  1%:  60.00%
```

```r
cat(sprintf("  5%%:  %.2f%%\n", nrej5  / nsim * 100))
```

```
  5%:  80.22%
```

```r
cat(sprintf("  10%%: %.2f%%\n", nrej10 / nsim * 100))
```

```
  10%: 87.58%
```

```r
cat("Rejection rates over 10000 simulations:\n")
```

```
Rejection rates over 10000 simulations:
```

```r
cat(sprintf("  1%%:  %.2f%%\n", Rnrej1  / nsim * 100))
```

```
  1%:  64.71%
```

```r
cat(sprintf("  5%%:  %.2f%%\n",Rnrej5  / nsim * 100))
```

```
  5%:  82.57%
```

```r
cat(sprintf("  10%%: %.2f%%\n", Rnrej10 / nsim * 100))
```

```
  10%: 88.84%
```

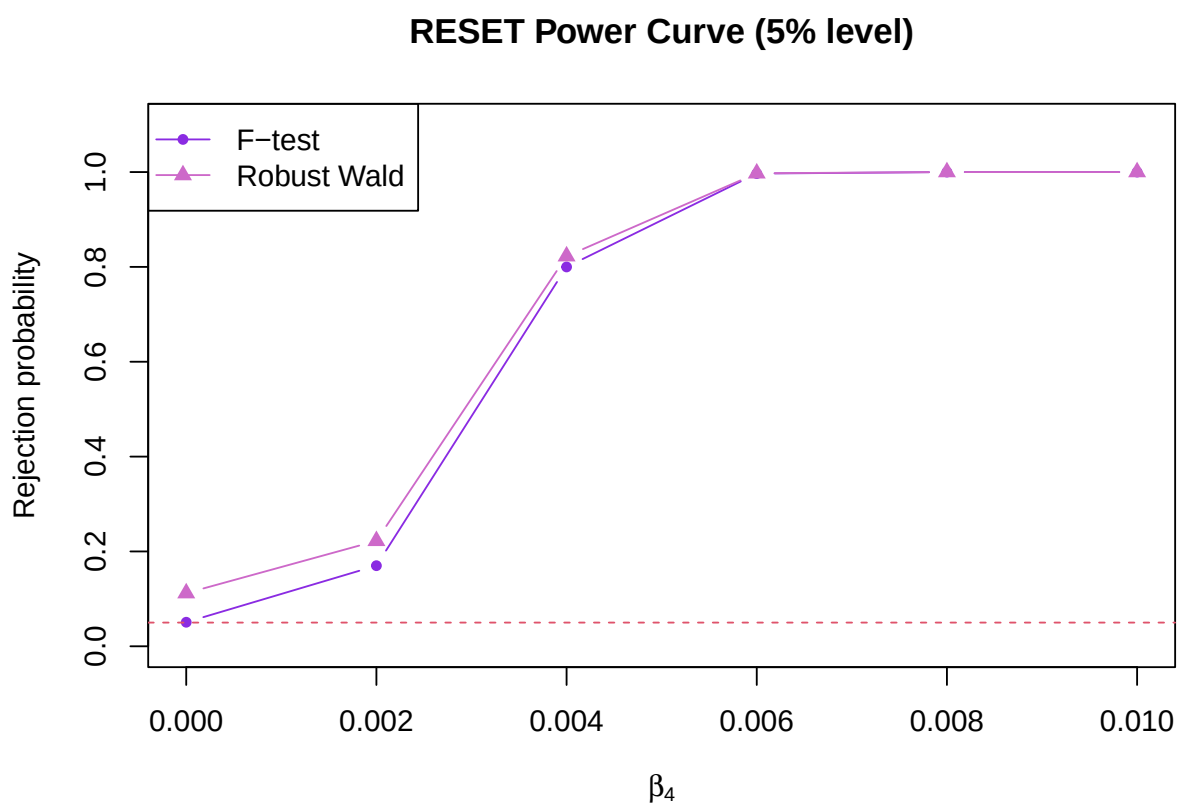
The null hypothesis is rejected very often, with at least 54.33% times with the robust test with significance level 0.01. At Significance 0.1, the hypothesis is rejected 85.95% of times with robust test and 87.58% with normal F-test.

Exercise 2.3: Power curve

We vary β_4 to see how the rejection rate changes. Later, we test for non-linearity using the RESET F-test and the robust Wald test and plot the rejection probability as a function of β_4.

```
b4_range=seq(0, 0.01, by = 0.002)
nsim2= 10000
q= 3                                    # number of restrictions (yhat^2, yhat^3, yhat^4)

rej_F= rep(0, length(b4_range))             # rejection counts for F
rej_W= rep(0, length(b4_range))             # rejection counts for robust Wald

F_cv_5= qf(0.95, q, n - 6)                  # critical value f 5%
W_cv_5= qchisq(0.95, q)                     # critical value chi 5%

R= cbind(matrix(0, q, 3), diag(q))       # constraint: coefficients = 0

set.seed(42)
for (j in 1:length(b4_range)) {
   beta0_j= c(1, 0.1, 0.025, b4_range[j])
   countF = 0
   countW = 0

   for (i in 1:nsim2) {
      eps = rnorm(n, 0, sigm0)
      ysim = Xdgp %*% beta0_j + eps

      # OLS on misspecified model
      bhat= XXinvX %*% ysim
      yhat= X %*% bhat
      e = ysim - yhat
      ssrR = sum(e * e)

      # RESET augmentation
      Xu  = cbind(X, yhat^2, yhat^3, yhat^4)
      bhat_u = solve(t(Xu) %*% Xu, t(Xu) %*% ysim)
      eu= ysim - Xu %*% bhat_u
      ssrU= sum(eu * eu)

      # Regular F-test
      Fstat  = ((ssrR - ssrU) / q) / (ssrU / (n - 6))
      if (Fstat > F_cv_5) countF = countF + 1

      # Robust Wald test
      XXinvU= solve(t(Xu) %*% Xu)
      ku= ncol(Xu)
      Sigma_hat = matrix(0, ku, ku)
      for (l in 1:n) {
         xi = Xu[l, ]
         Sigma_hat = Sigma_hat + eu[l]^2 * xi %*% t(xi)
      }
      Var_rob = XXinvU %*% Sigma_hat %*% XXinvU
      Rb_hat  = R %*% bhat_u
      Wrob  = t(Rb_hat) %*% solve(R %*% Var_rob %*% t(R)) %*% Rb_hat
```

```r
      if (Wrob > W_cv_5) countW = countW + 1
   }
   rej_F[j] = countF / nsim2
   rej_W[j] = countW / nsim2
}

# Plot both power curves
plot(b4_range, rej_F, type = "b", pch = 20, col = "blueviolet",
     xlab = expression(beta[4]), ylab = "Rejection probability",
     main = "RESET Power Curve (5% level)", ylim = c(0, max(rej_F, rej_W) * 1.1))
lines(b4_range, rej_W, type = "b", pch = 17, col = "orchid3")
abline(h = 0.05, col = 2, lty = 2)
legend("topleft", legend = c("F-test", "Robust Wald"),
       col = c("blueviolet", "orchid3"), pch = c(20, 17), lty = 1)
```




```r
cat("Power curve values (5% level):\n\n")
```

```
Power curve values (5% level):
```

```r
cat(sprintf("  %-10s  %10s  %12s\n", "beta4", "F-test", "Robust Wald"))
```

```
  beta4           F-test   Robust Wald
```

```r
for (j in 1:length(b4_range)) {
   cat(sprintf("  %-10.3f  %9.2f%%  %11.2f%%\n",
       b4_range[j], rej_F[j] * 100, rej_W[j] * 100))
}
```

```
  0.000            5.07%        11.20%
  0.002           17.00%        22.23%
  0.004           79.99%        82.25%
  0.006           99.68%        99.74%
  0.008          100.00%       100.00%
  0.010          100.00%       100.00%
```

At $\beta_4 = 0$ the simulated rejection probability should be near 5%. As β_4 increases, omitting exp^2 causes more difference between the true model and the omitted one. The RESET test in that case states about non-linearity more often, so the rejection rises. Power curve shows us how the probability of "correctly" rejecting H_0 increases with the size of the omitted variable effect.

Exercise 2.4: Heteroscedastic errors

We introduce heteroscedastic errors: $\varepsilon_i \sim N(0,\ \exp(\alpha \cdot educ_i))$ with $\alpha = 0.1$.

```
alpha_het = 0.1
nsim3     = 10000

cv_F_1  = qf(0.99, 3, n - 6)
cv_F_5  = qf(0.95, 3, n - 6)
cv_F_10 = qf(0.90, 3, n - 6)
cv_bp_1  = qf(0.99, 4, n - 5)
cv_bp_5  = qf(0.95, 4, n - 5)
cv_bp_10 = qf(0.90, 4, n - 5)

# Counters for RESET
nrej_reset_1 = 0; nrej_reset_5 = 0; nrej_reset_10 = 0
# Counters for BP
nrej_bp_1 = 0; nrej_bp_5 = 0; nrej_bp_10 = 0

set.seed(999)
for (i in 1:nsim3) {
   # Heteroscedastic errors: Var(eps_i) = exp(alpha * educ_i)
   sd_i = exp(alpha_het * educ / 2)     # sd so that Var = exp(alpha * educ)
   eps  = rnorm(n, 0, sd_i)
   ysim = Xdgp %*% beta0 + eps

   bhat   = XXinvX %*% ysim                     # OLS on misspecified model
   e      = ysim - X %*% bhat
   ssrR   = sum(e * e)

   # RESET test
   yhat   = X %*% bhat
   Xu     = cbind(X, yhat^2, yhat^3, yhat^4)
   bhat_u = solve(t(Xu) %*% Xu, t(Xu) %*% ysim)
   eu     = ysim - Xu %*% bhat_u
   ssrU   = sum(eu * eu)
   Fstat  = ((ssrR - ssrU) / 3) / (ssrU / (n - 6))

   if (Fstat > cv_F_1)  nrej_reset_1  = nrej_reset_1  + 1
   if (Fstat > cv_F_5)  nrej_reset_5  = nrej_reset_5  + 1
   if (Fstat > cv_F_10) nrej_reset_10 = nrej_reset_10 + 1

  # Testing for heteroscedasticity
  # We create regression of different combinations of residuals (Using White's statistic) and check if

  X_bp = cbind(rep(1, n),educ,exp,educ^2,exp^2)

  gamma_hat = solve(t(X_bp) %*% X_bp, t(X_bp) %*% e^2) # We use e, which is residuals from restricted m
  e_bp = e^2 - X_bp %*% gamma_hat
  SSR_bp = sum(e_bp*e_bp)
  SST_bp = sum((e^2-mean(e^2))^2)
  R2_bp= (1-(SSR_bp/SST_bp))
  Fstat_bp  = (R2_bp / 4) / ((1-R2_bp) / (n - 5)) #Fstat where we have p-1=4 features (including the sq
```

```r
  if (Fstat_bp > cv_bp_1)  nrej_bp_1  = nrej_bp_1  + 1
  if (Fstat_bp > cv_bp_5)  nrej_bp_5  = nrej_bp_5  + 1
  if (Fstat_bp > cv_bp_10) nrej_bp_10 = nrej_bp_10 + 1




}

cat("With heteroscedastic errors (alpha = 0.1):\n\n")
```

```
With heteroscedastic errors (alpha = 0.1):
```

```r
cat("RESET test rejection rates:\n")
```

```
RESET test rejection rates:
```

```r
cat(sprintf("  1%%: %.2f%%   5%%: %.2f%%   10%%: %.2f%%\n",
    nrej_reset_1 / nsim3 * 100, nrej_reset_5 / nsim3 * 100,
    nrej_reset_10 / nsim3 * 100))
```

```
  1%: 8.85%   5%: 23.86%   10%: 34.83%
```

```r
cat("\nBreusch-Pagan test rejection rates:\n")



Breusch-Pagan test rejection rates:
```

```r
cat(sprintf("  1%%: %.2f%%   5%%: %.2f%%   10%%: %.2f%%\n",
    nrej_bp_1 / nsim3 * 100, nrej_bp_5 / nsim3 * 100,
    nrej_bp_10 / nsim3 * 100))
```

```
  1%: 67.60%   5%: 84.73%   10%: 90.73%
```

With heteroscedastic errors, the Breusch–Pagan test should reject H_0 much more frequently than the nominal level, confirming its power against heteroscedasticity. The RESET test may also show elevated rejection rates since heteroscedasticity inflates the F-statistic when regular S.E. are used.